



# JORGE LEAL

Software Engineering Student &  
Data Analysis Enthusiast

773-178-3694  
jlealcornejo@gmail.com  
@jorgelealdev.com

## PROJECTS

### Agent System | TypeScript + VS Code API + SQLite + LLM Orchestration

VS Code extension implementing a project-aware multi-agent coding assistant with persistent memory, optional RAG, and secure tool execution integrated into the Sidebar and Copilot Chat (@agent).

- Clean Architecture & monorepo: decoupled core (EventBus, orchestration, interfaces), infra adapters, and app activation layer in pnpm workspaces.
- Specialized agents & coordination: Coder, Reviewer, Tester, and Coordinator with hybrid routing (keywords + embeddings + LLM), TTL caching, agent collaboration, and Auto/Agent/Plan modes with configurable reasoning levels.
- Advanced memory & RAG: persistent SQLite memory (session/project/global) with TTL, optional OpenAI embeddings, hash-gated indexes, and static workspace analysis (AI Project Awareness / Project Context).
- Secure tool management & LLM fallback: read/edit/create/delete files, run\_tests, and search\_web with risk-based approval; chained LLMs (Copilot → VS Code LM → MCP fallback); configurable web search provider.

Stack: TypeScript (ES2022), VS Code API ^1.99, SQLite (wa-sqlite, sql.js), Groq SDK, OpenAI, MCP, esbuild, pnpm workspaces, zod, DOMPurify.

### PixMinder | Rust (Axum) + PostgreSQL + Redis

Intelligent image optimization service designed as a high-performance CDN alternative to replace Cloudinary native transformations.

- Backend built in Rust with Axum following Clean / Hexagonal Architecture and strict layer separation.
- Context-aware image optimization with automatic format selection (AVIF, WebP, JPEG, PNG), adaptive quality, DPR-aware resizing, and SVG pass-through.
- Multi-layer caching system (in-memory, PostgreSQL, Redis) achieving sub-10ms cache latency and 15–30% better compression than Cloudinary native.
- Currently serving production traffic for Qubly CRM and LaZalza with +7–13 Lighthouse Performance points and 300–600ms LCP reduction.

Stack: Rust, Axum, Tokio, PostgreSQL (Neon), Redis (Upstash), SQLx, Moka, MozJPEG, WebP, AVIF (ravif), Cloudinary, Chart.js, Docker, Fly.io.

### Qubly CRM | SvelteKit + Go (Fiber) + PostgreSQL (Neon)

Multi-tenant SaaS platform for small businesses in Mexico to launch an online store in minutes with direct WhatsApp ordering and zero sales commissions

- Hybrid architecture with SvelteKit (SSG + SSR + PWA) frontend and Go backend following Clean Architecture principles.
- Full e-commerce system with recurring subscriptions, Stripe Connect, webhooks, and a built-in referral system.
- AI-powered chatbot using Groq (Llama) optimized for low cost via intent detection, efficient context construction, and response caching.

Stack: TypeScript, SvelteKit, Go, Fiber, PostgreSQL (Neon), Stripe, Groq AI, Tailwind CSS, Docker, Fly.io.

### Kiddyboard | SvelteKit + FastAPI + PostgreSQL (Neon)

Multi-role management system for children's entertainment centers with administrative dashboard, sales, inventory, services, and real-time reporting.

- SvelteKit frontend with hybrid rendering and monorepo architecture; FastAPI backend following Clean Architecture with serverless PostgreSQL.
- Operational dashboard with real-time metrics (sales, ATV, inventory, active timers), comparative reports, and statistical forecasting for sales and capacity planning.
- Advanced features include real-time timers, Excel/PDF report exports, digital menu generation (PDF & video), and granular role-based access control.

Stack: TypeScript, SvelteKit, FastAPI, PostgreSQL (Neon), Chart.js, Tailwind CSS, Docker, Fly.io.

### LaZalza | Next.js + KeystoneJS + PostgreSQL

News and editorial website about music, cinema, and popular culture.

- Frontend with Next.js 14 App Router and ISR (5 min).
- Headless CMS with KeystoneJS + PostgreSQL.
- Cloudinary integration and interactive 3D logo.
- Stack: TypeScript, React, Tailwind CSS, GraphQL, Docker, Cloudinary

### Raccoonin Records | Next.js + KeystoneJS + PostgreSQL

Platform for an independent music label with hybrid architecture.

- Migrated from CSR to ISR and posts with infinite pagination.
- Multimedia embeds, streaming metalinks, and dynamic sitemap.

Stack: TypeScript, Next.js 14, React, Tailwind CSS, GraphQL, KeystoneJS, Cloudinary, Three.js, Docker

### Beat Catalogue | SvelteKit + FastAPI + PostgreSQL

Music catalog with hybrid SSR and Spotify-style modal overlay.

- Backend with FastAPI + PostgreSQL and custom admin panel.
- Audio player with shuffle, play counter, and persistent theme switcher.

Stack: TypeScript, SvelteKit, Tailwind CSS, FastAPI, SQLAlchemy, Cloudinary, Three.js

### Databoard | FastAPI + NiceGUI + Plotly + PostgreSQL (Neon)

Multi-tenant analytics dashboard for real-time monitoring of web metrics across multiple sites and projects.

- Async architecture with tenant-level data isolation, optimized caching, and resilient error handling.
- Advanced Plotly visualizations: time-series charts, hourly heatmaps, geolocation maps, and Spotify-style KPIs.
- Key features include multi-site selection, audience breakdown, time zone adjustment, CSV export, and statistical metric forecasting (EWMA, trend analysis).

Stack: Python, FastAPI, NiceGUI, Plotly, PostgreSQL (Neon), Tailwind CSS, Docker, Fly.io.

### BSTCGenerator | FastAPI + MusicGen + Gemini

Audio generation platform for producers with cost-efficient architecture.

- Decoupled FastAPI backend with AI worker running on Google Colab (GPU) using MusicGen.
- Embedding and recommendation system based on feedback and pattern analysis.

Stack: Python, FastAPI, SQLite, Librosa, MusicGen, Gemini, Vanilla JS, GitHub Codespaces, Google Colab.

## PROFILE

Fullstack Developer focused on web platforms, high-performance backend systems, and data-driven dashboards.

Experience building multi-tenant SaaS products, hybrid (SSG/SSR) architectures, headless CMS solutions, and performance-optimized services leveraging advanced caching strategies.

Hands-on experience designing LLM-powered tools, multi-agent orchestration systems, and clean architectures for development workflows and automation.

## LANGUAGES

- Spanish: Native
- English: Intermediate

## EDUCATION

### Universidad Autónoma Metropolitana, Unidad Xochimilco

Bachelor's in Social Communication

Graduated: October 2015

### Universidad Abierta y a Distancia de México

Bachelor's in Software Engineering

Started: January 2025 | Currently studying

## STACK

- **Frontend:** Next.js 14, React 18, SvelteKit, Tailwind CSS, NiceGUI, Astro, Three.js
- **Backend:** Rust (Axum), Go (Fiber), KeystoneJS, Node.js, Python
- **Databases:** PostgreSQL (Neon), SQLite, Redis
- **AI / LLM Systems:** OpenAI, Groq, Gemini, MusicGen, VS Code Language Model API, MCP
- **DevOps / Deploy:** Docker, Fly.io, Cloudflare, GitHub Codespaces
- **Architecture & Tooling:** Clean Architecture, Ports & Adapters, Monorepos (pnpm workspaces), Event-Driven Systems
- **Media & Performance:** Cloudinary, proprietary image CDN (PixMinder)
- **Others:** Stripe, GraphQL (URQL), Axios, Plotly